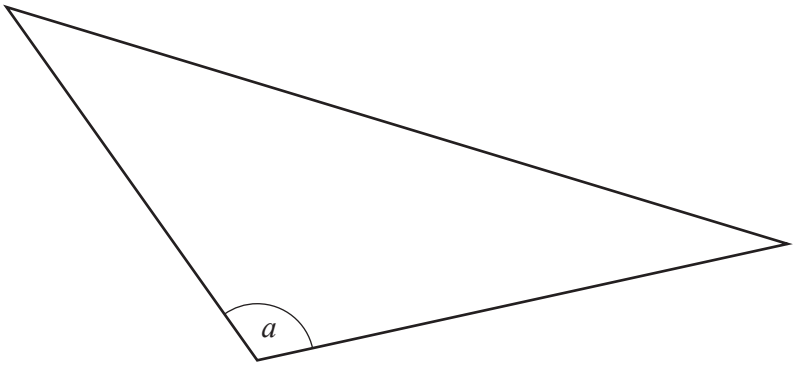
0580_s22_qp_32_Q3

3 **(a)**



(i) Write down the mathematical name for the type of angle *a*.

..... [1]

(ii) Measure angle *a*.

..... [1]

(b) Kate describes a quadrilateral.

- All the sides are the same length.
- It has only two lines of symmetry.

(i) Draw a sketch of this quadrilateral.

[1]

(ii) Write down the mathematical name for this quadrilateral.

..... [1]

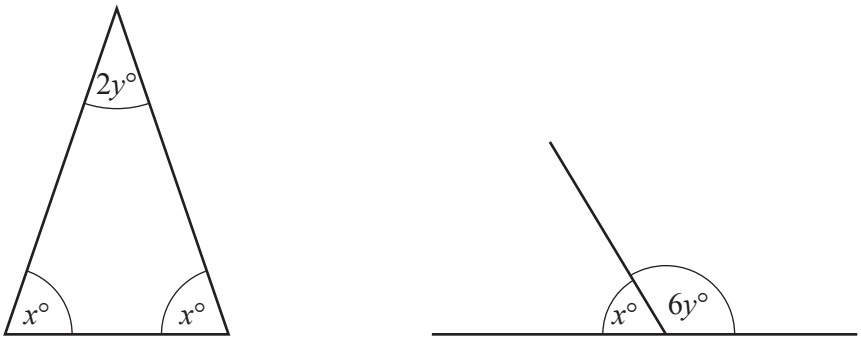
(iii) One of the interior angles of this quadrilateral is 70°.

Work out the other three interior angles.

..... , ... , ... [2]

0580_s22_qp_32_Q3

(c) The diagrams show the angles in a triangle and two angles on a straight line.



NOT TO SCALE

(i) The triangle is used to write down an equation in terms of x and y.

$$2x + 2y = 180$$

Give the geometrical reason why this equation is correct.

Reason ..... [1]

(ii) Use the diagram with two angles on a straight line to write down another equation in terms of x and y.

..... [1]

(iii) Solve these simultaneous equations.
You must show all your working.

$x =$ .....

$y =$ ..... [3]